
Fast Tree Diffusions for Probabilistic Tabular Regression via Flow Matching

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Abstract

1 We extend Treeffuser, a score-based diffusion model for probabilistic tabular
2 regression with gradient-boosted trees, in two directions. A score-side recipe—
3 conditional-mean residualization, EDM-style preconditioning, log-sigma time sam-
4 pling, and noise-level features—improves aggregate calibration after per-dataset
5 tuning, reducing 95% coverage error from 0.023 to 0.015 while cutting sampling
6 time $6.5\times$ over the published baseline. A flow-matching variant with Gaussian
7 paths and algebraic score recovery gives the best normalized CRPS among Treef-
8 fuser rows with a VP path and deterministic five-step Heun ODE, and is another
9 $2.4\times$ faster than the improved score model, but gives back some coverage. The
10 common mechanism is better conditioning of the supervised regression problem
11 seen by the tree ensemble. Under this ten-dataset tuning protocol, the Treeffuser
12 variants give the best aggregate normalized CRPS and coverage metrics among
13 displayed rows, though gains over the published baseline are directional rather than
14 conventionally significant ($p \approx 0.07$), and tuned non-diffusion baselines still win
15 several datasets on raw CRPS. Stochastic ($\varepsilon > 0$) samplers do not Pareto-dominate
16 the deterministic corner, so the empirical recommendation is $\varepsilon = 0$ when aggregate
17 CRPS and latency are the target, while stochasticity remains a conditional-tail
18 calibration knob.

19 1 Introduction

20 Probabilistic regression on tabular data must combine the predictive strength of gradient-boosted
21 trees with full predictive distributions usable for calibration, decision-making, and uncertainty
22 quantification. Treeffuser [Beltrán-Velez et al., 2024] achieves this by training LightGBM [Ke et al.,
23 2017] regressors to estimate the score of a noise-perturbed conditional density and integrating the
24 corresponding reverse-time stochastic differential equation (SDE) [Song et al., 2021] at inference. The
25 framework inherits the calibration properties of score-based diffusion while avoiding neural-network
26 density estimators.

27 Our notion of tractability is computational rather than likelihood-exact: predictive quantities are
28 Monte Carlo estimates, but training reduces to supervised tree regression, the final FM row samples
29 with a five-step Heun ODE, and Eq. (2) recovers scores when stochastic sampling is desired.

30 We develop Treeffuser along two axes: score-side modifications matched to histogram-bin tree
31 learners, and Gaussian-path flow matching [Lipman et al., 2023, Liu et al., 2023], which removes the
32 indirect score-to-drift mapping and opens access to stochastic interpolants [Albergo et al., 2023] with
33 non-negative stochasticity schedule $\varepsilon(t)$.

34 **Contributions.** We make three contributions:

1. *Score-side modifications for tree-based diffusion*: conditional-mean residualization, EDM preconditioning, log- σ time sampling, and an explicit log- σ feature for LightGBM score regression (§2.1).
2. *Flow matching for Treeffuser*: a Gaussian-path FM trainer with score recovery when stochastic-interpolant sampling or endpoint-error analysis requires it (§2.2, Lemma 1).
3. *Regression-conditioning unification*: regression conditioning for the tree ensemble as the bottleneck behind both score- and flow-side gains (§4, Appendix A).

Across ten UCI benchmarks, with one fold for tuning and five for evaluation, the score-side recipe gives the best aggregate calibration and CRPS skill score among Treeffuser rows, while FM gives the best aggregate normalized CRPS and fastest Treeffuser sampler. The shared lesson is that noising path, parameterization, training distribution, features, and sampler must make the induced supervised problem well matched to histogram splits.

2 Method

2.1 Score-Side Modifications

The published Treeffuser uses VESDE marginals with uniform t sampling, a noise-prediction target, and LightGBM features $[y_t, X, t]$. We change four pieces: response/input conditioning, allocation of training rows over noise levels, explicit noise-level features, and sampler efficiency.

Conditional-mean residualization. We fit a cross-validated LightGBM mean predictor $\hat{\mu}(x)$ and apply diffusion to $y - \hat{\mu}(x)$ rather than y , moving conditional location out of the score-regression target. The residualizer uses a high-capacity configuration selected from a prior diagnostic sweep (Appendix D) and then fixed; it is not tuned jointly with the diffusion or flow hyperparameters. On large datasets this fixed residualizer can become a bottleneck once the inner model has enough data to represent the conditional mean directly.

EDM preconditioning. Following Karras et al. [2022], we predict a denoised target $D_\theta(y_t, t)$ scaled to unit variance across noise levels and reconstruct the score $s(y_t, t) = (D_\theta - y_t)/\sigma(t)^2$. This stabilizes the regression target for trees, which would otherwise need to extrapolate noise-prediction magnitudes across t .

Log-sigma time sampling. We draw $\log \sigma(t) \sim \mathcal{N}(-1.2, 1.2^2)$, clip to the SDE’s achievable range, and invert. Because trees bin features by data density, the training-time distribution directly controls where the score model spends capacity.

Log-sigma noise feature. The feature vector for LightGBM becomes $[y_t, X, t, \log \sigma(t)]$, giving the regressor an explicit noise-level signal aligned with EDM scaling.

2.2 Flow Matching with Gaussian Paths

A Gaussian flow path interpolates data y_0 and a standard-normal prior z via

$$y_t = \alpha(t) y_0 + \beta(t) z, \quad t \in [0, 1], \quad (1)$$

with boundary conditions $\alpha(0) = 1, \alpha(1) = 0, \beta(0) = 0, \beta(1) = 1$. We compare three paths: *linear* ($\alpha = 1 - t, \beta = t$), *trigonometric* ($\alpha = \cos(\pi t/2), \beta = \sin(\pi t/2)$), and a *VP* schedule with linear β_t giving $\alpha^2(t) = \exp(-T(t)), T(t) = \frac{1}{2}\beta_{\min}t + \frac{1}{4}(\beta_{\max} - \beta_{\min})t^2$. The training target is the conditional velocity $u_t = \alpha'(t)y_0 + \beta'(t)z$, regressed against the model’s $v_\theta(y_t, x, t)$.

Velocity-to-score formula. For any Gaussian path with Wronskian $W(t) = \alpha(t)\beta'(t) - \alpha'(t)\beta(t)$, the conditional score is algebraically recoverable from the population velocity:

$$\nabla \log p_t(y_t | x) = \frac{\alpha'(t) y_t - \alpha(t) v_\theta(y_t, x, t)}{W(t) \beta(t)}. \quad (2)$$

For linear FM this reduces to $-(y_t + (1 - t)v)/t$; for trig to $-y_t - (2/\pi) \cot(\pi t/2)v$. We use this identity operationally: it lets a velocity-trained Treeffuser enter the same stochastic-interpolant sampler family as score matching and exposes the endpoint amplification studied in Appendix A.

78 **Stochastic-interpolant family.** Albergo et al. [2023] show that any non-negative $\varepsilon(t)$ defines a
 79 marginal-preserving reverse-time sampler at the population velocity/score:

$$dy_\tau = \left(-v_\theta + \frac{\varepsilon^2}{2} s_\theta\right) d\tau + \varepsilon dW_\tau, \quad (3)$$

80 where $\tau = 1 - t$ is reverse time and s_θ is the score from (2). The choice $\varepsilon \equiv 0$ recovers the
 81 deterministic probability-flow ODE; positive $\varepsilon(t)$ broadens samples, with exact marginal preservation
 82 only for the true velocity/score. We use the schedule $\varepsilon(t) = ct$, which vanishes at the data endpoint
 83 and suppresses recovered-score error in the stochastic drift (see Proposition 2).

84 **Why these choices suit trees.** Eq. (2) makes FM compatible with stochastic-interpolant samplers;
 85 variance-preserving paths keep the tree input y_t near unit scale, unlike a linear path with $\text{Var}(y_t) \approx$
 86 $(1 - t)^2 + t^2$; and $\varepsilon = 0$ avoids recovered-score error in the stochastic drift (Appendix A). The path
 87 choice is therefore a finite-regression and sampler tradeoff, not a theorem that VP must dominate
 88 every tuned linear-flow row.

89 2.3 Implementation Notes

90 The FM code reuses Treeffuser’s per-output-dimension LightGBM fitter and feature pipeline; only
 91 the target and prior change. Sampling uses the same Heun integrator and residualizer as the score
 92 path, and both training and sampling avoid the singular endpoint with $t \geq 10^{-5}$.

93 3 Experiments

94 **Datasets and protocol.** Ten UCI benchmarks (Table 2) are evaluated with a nested tuning protocol.
 95 We split each dataset into six folds: fold 0 is used only for hyperparameter selection, and folds 1–5
 96 are evaluation folds. For each dataset/model family, Optuna tunes on the fold-0 train/validation split;
 97 the selected configuration is then refit and evaluated on each held-out evaluation fold. Evaluation-fold
 98 training rows range from 256 on yacht to 38,108 on protein. For sample-based methods, each test
 99 point uses 200 samples; Treeffuser sampler choices are detailed in Appendix B. Metrics are averaged
 100 over evaluation folds and, for headline summaries, over datasets.

101 **Metrics.** We report CRPS skill score $\text{CRPSS} = 1 - \text{CRPS}_{\text{model}}/\text{CRPS}_{\text{climatology}}$, mean rel-CRPS,
 102 DSS [Dawid, 1984], PIT-KS pass rate at $\alpha = 0.05$, absolute coverage error at 50/90/95% nominal
 103 levels [Gneiting and Raftery, 2007], and sample-generation time. CRPSS and rel-CRPS are the
 104 aggregate comparison metrics because raw CRPS is target-scale dependent; mean climatological
 105 CRPS ranges from 0.0043 on naval to 44.8 on diabetes. Timings are measured on one Apple-
 106 arm64 workstation, exclude fitting, and include generating 200 samples per test point plus inverse
 107 transformations. They use practical benchmark settings rather than a single-thread cap: Treeffuser
 108 samples in batches with `n_parallel=20` and LightGBM `n_jobs=-1`; other libraries use their
 109 configured parallel defaults.

110 **Variants compared.** Three Treeffuser rows compare bundled recipes: (1) PUBLISHED—the original
 111 baseline with noise-prediction, uniform t , no residualization; (2) SCORE+—the baseline plus §2.1;
 112 (3) FM—VP flow matching with residualization and a 5-step Heun ODE. Each row is tuned per
 113 dataset under the same fold-0 protocol. The residualizer-C configuration used in the headline rows
 114 was selected from the prior diagnostic sweep in Appendix D and then fixed; it is not tuned jointly with
 115 the diffusion or flow hyperparameters. We also compare against six per-dataset tuned probabilistic
 116 baselines: NGBoost [Duan et al., 2020], quantile-regression LightGBM [Ke et al., 2017], CatBoost
 117 RMSE-with-uncertainty [Prokhorenkova et al., 2018], iBUG [Brophy and Lowd, 2022], a Gaussian
 118 deep ensemble [Lakshminarayanan et al., 2017], and CARD-style neural diffusion [Han et al., 2022].

119 3.1 Headline Result

120 Table 1 reports cross-dataset means under the tuning/evaluation protocol. The score-side recipe lifts
 121 CRPSS from 0.670 to 0.681, raises PIT-uniformity pass rate from 0.40 to 0.62, reduces 90/95%
 122 coverage error from 0.044/0.023 to 0.020/0.015, and cuts Treeffuser sample time $6.5\times$. FM is
 123 close on CRPSS, has the best aggregate normalized CRPS among displayed rows, and is another

Table 1: Headline real-data results, mean over 10 UCI benchmarks with fold 0 used for tuning and folds 1–5 used for evaluation. rel-CRPS is the per-dataset CRPS divided by the best displayed variant on that dataset. $|cE|@q$ is absolute coverage error at the $q\%$ interval level. The CARD row is a compact benchmark adapter, not a fully optimized CARD pipeline (see Appendix H).

Variant	CRPSS $\uparrow$	rel-CRPS $\downarrow$	KS $p > .05 \uparrow$	$ cE @50 \downarrow$	$ cE @90 \downarrow$	$ cE @95 \downarrow$	samp. time
Treeffuser-published	0.670	1.242	0.40	0.117	0.044	0.023	221.35 s
Treeffuser-score ⁺ (ours)	0.681	1.131	0.62	0.069	0.020	0.015	34.12 s
Treeffuser-FM (ours)	0.680	1.117	0.46	0.064	0.028	0.024	14.41 s
QReg-LightGBM	0.668	1.459	0.34	0.066	0.041	0.044	0.30 s
CatBoost-unc.	0.638	1.409	0.28	0.093	0.078	0.066	0.01 s
NGBoost	0.601	1.550	0.16	0.119	0.114	0.101	0.04 s
Deep ensemble	0.654	1.519	0.26	0.090	0.032	0.018	0.03 s
iBUG	0.649	1.471	0.36	0.111	0.051	0.065	2.89 s
CARD-style diffusion	0.631	1.632	0.24	0.086	0.057	0.043	42.84 s

Table 2: Per-dataset CRPSS (higher is better) and post-split training rows.

Dataset	Train rows	Published	Score ⁺	FM
yacht	256	0.941	0.962	0.962
diabetes	368	0.201	0.241	0.226
energy	640	0.956	0.962	0.964
concrete	858	0.740	0.774	0.771
wine	5414	0.346	0.305	0.307
kin8nm	6826	0.555	0.617	0.631
power	7973	0.829	0.839	0.843
naval	9945	0.950	0.949	0.951
cali. housing	17200	0.687	0.676	0.674
protein	38108	0.494	0.480	0.472

2.4 $\times$ faster than score⁺, but gives back part of score⁺’s coverage advantage. Per-dataset raw-CRPS winners span five families; a Treeffuser variant wins on 6/10 datasets, with the remaining wins split across deep ensembles, CARD-style diffusion, and quantile-regression LightGBM. The Treeffuser aggregate advantage is directional rather than conventionally significant: paired Wilcoxon tests give one-sided $p = 0.080$ for score⁺ over published and $p = 0.065$ for FM over published; FM versus score⁺ remains indistinguishable ($p = 0.922$, two-sided).

3.2 Per-Dataset CRPSS

Table 2 shows the Treeffuser per-dataset CRPSS. Score⁺ is best on concrete and diabetes, FM on energy, kin8nm, naval, power, and yacht, and the published baseline on wine and the two largest datasets. Thus the tuned comparison exposes a Pareto surface: score⁺ is usually the calibrated choice, FM the fast normalized-CRPS choice, and the published SDE remains competitive when full SDE sampling is affordable.

3.3 Robustness Checks

Table 3 summarizes the same tuned artifacts by per-dataset ranks and dataset-level standard errors. The Treeffuser rows occupy the two best mean CRPS ranks, but the aggregate gaps are small relative to the dataset standard errors, and the rank tests do not support a one-family sweep. A Friedman test over the nine displayed families rejects equal CRPS ranks ($p = 0.001$), but the Nemenyi critical difference is 3.80 mean-rank points; post-hoc gaps separate FM and score⁺ only from the weakest rows, not from the published SDE, quantile-regression LightGBM, or deep ensembles. This supports the intended reading of Table 1: the aggregate Treeffuser gains are robustly directional, while individual datasets and metrics still have mixed winners.

3.4 Mechanism Ablation

Table 4 isolates the main design choices under the same fold-0 tuning protocol and the same LightGBM hyperparameter search surface. Three conclusions matter for the mechanism story. First,

Table 3: Robustness summary from the tuned evaluation artifacts. Ranks average per-dataset ranks after first averaging folds 1–5; lower ranks are better. SE is the standard error over the ten dataset-level CRPS means.

Model	CRPS rank	CRPS $\pm$ SE	rel-CRPS	cE @95 rank
Treeffuser-FM	2.80	0.680 $\pm$ 0.086	1.117	4.00
Treeffuser-score ⁺	3.00	0.681 $\pm$ 0.085	1.131	2.10
QReg-LightGBM	4.00	0.668 $\pm$ 0.079	1.459	5.60
Treeffuser-published	4.10	0.670 $\pm$ 0.084	1.242	3.30
Deep ensemble	5.80	0.654 $\pm$ 0.082	1.519	2.90
iBUG	5.80	0.649 $\pm$ 0.081	1.471	6.50
CatBoost-unc.	6.00	0.638 $\pm$ 0.088	1.409	7.10
CARD-style diffusion	6.60	0.631 $\pm$ 0.091	1.632	5.70
NGBoost	6.90	0.601 $\pm$ 0.094	1.550	7.80

Table 4: Tuned mechanism ablation over the same ten UCI datasets. Each row uses the same LightGBM hyperparameter search surface and fold-0 tuning / folds-1–5 evaluation protocol; fixed method choices and bound samplers differ by row. rel-CRPS is normalized by the best ablation row on each dataset.

Variant	CRPS $\uparrow$	rel-CRPS $\downarrow$	q -MACE $\downarrow$	cE @90 $\downarrow$	cE @95 $\downarrow$	samp. time	raw wins
Score noise, Euler-50	0.670	1.146	0.044	0.044	0.023	217.30 s	2
Score noise, Heun-25	0.629	1.418	0.058	0.049	0.025	190.92 s	0
+ residualizer	0.646	1.101	0.041	0.044	0.027	47.58 s	0
+ EDM input/target	0.680	1.039	0.034	0.025	0.016	109.46 s	0
+ log- σ feature	0.679	1.040	0.035	0.027	0.017	73.10 s	1
Score ⁺ full	0.681	1.042	0.028	0.020	0.015	35.11 s	0
FM linear + residualizer	0.684	1.024	0.028	0.024	0.020	19.28 s	4
FM VP, no residualizer	0.631	1.975	0.050	0.051	0.032	23.19 s	1
FM VP + residualizer	0.680	1.031	0.034	0.028	0.024	14.69 s	2

changing only the published noise-prediction score model from Euler SDE sampling to a 25-step Heun probability-flow ODE is not a free speedup: CRPS falls from 0.670 to 0.629. Second, EDM input/target preconditioning is the largest score-side jump, improving CRPS from 0.646 after residualization alone to 0.680 and cutting 95% coverage error from 0.027 to 0.016. The log- σ feature is roughly neutral under uniform t , while the full score⁺ bundle is the most calibrated score-side row. Third, few-step FM needs residualization: VP-FM without residualization is the weakest rel-CRPS row despite one small-dataset win. The tuned linear-FM residualized row has the best aggregate CRPS/rel-CRPS in this ablation, while VP-FM remains the faster FM row. This weakens a simple “VP geometry always wins” reading, but supports the broader finite-regression view: feature scale, velocity target shape, residualization, and ODE discretization all affect the supervised problem solved by LightGBM.

3.5 Sampler Cost Tradeoff

Figure 1 re-evaluates the tuned Treeffuser configurations across sampler step counts without retuning hyperparameters. The practical pattern is sharp. FM-ODE is effectively converged by 3–5 Heun steps: VP-FM moves from rel-CRPS 1.046 at 3 steps to 1.037 at 5 steps, and residualized linear FM is already at 1.031 at 3 steps. Score⁺ needs more integration: 5-step PF-ODE is not competitive (rel-CRPS 1.781), while 15–25 steps recover the calibrated score row. The published Euler SDE improves gradually with step count but remains much slower at comparable aggregate CRPS. Finally, the 25-step FM-SDE points confirm the calibration tradeoff: for VP-FM, stochasticity improves 95% coverage error from 0.024 to 0.018, but worsens CRPS, q -MACE, and 90% coverage error. Thus the default recommendation remains few-step FM-ODE for aggregate CRPS/latency, with stochasticity reserved for conditional-calibration targets.

Design diagnostics. Fixed diagnostics explain the bundles but are not separately tuned factorial comparisons. For FM, the fixed path sweep favours VP under matched residualizer-C and a 5-step Heun ODE, while the tuned ablation in Table 4 shows that residualized linear FM can be the stronger aggregate CRPS row. Thus we treat path choice as a practical conditioning/sampling tradeoff rather than a settled hierarchy. Stochastic samplers do not Pareto-dominate $\varepsilon = 0$ on aggregate CRPS/coverage/latency, though they can improve high-IQR tail coverage (Appendix K); transferred

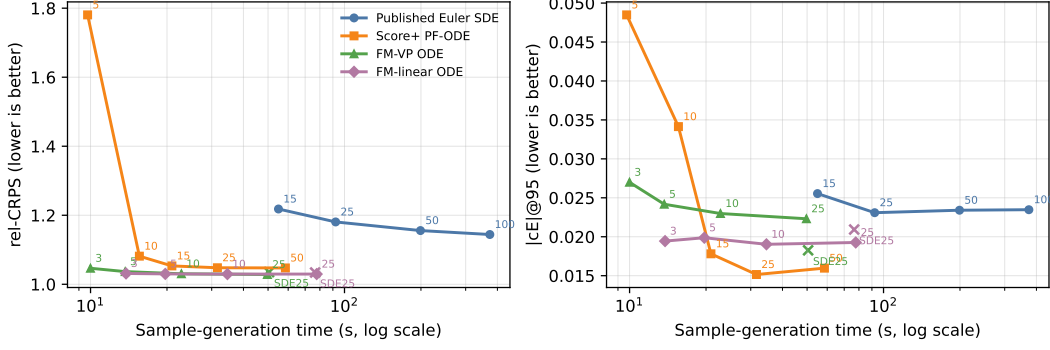


Figure 1: Sampler cost tradeoff after tuning, averaged over the same ten UCI datasets and folds 1–5. Points are eval-only sampler changes from the tuned configurations; labels show solver steps, and SDE25 marks the 25-step stochastic FM sampler. rel-CRPS is normalized by the best sampler-cost row on each dataset.

score-side tricks are neutral or worse. This is why the recommendation is deterministic ODE for aggregate CRPS/latency, with stochasticity left as a conditional-calibration knob. Appendix C maps tuned headline claims versus fixed diagnostics.

4 Discussion

Two findings deserve emphasis. First, the headline rows compare tuned bundled recipes, while Appendix C scopes the fixed diagnostics. Because score⁺ and FM share the same LightGBM tuning surface, the FM result is not implicitly absorbing tuning-budget improvements that belong to the score track. Second, per-dataset tuning turns the headline into a tradeoff: score⁺ is the strongest calibrated Treeffuser row, whereas FM is strongest on normalized CRPS and latency. The common mechanism is regression conditioning for tree ensembles: EDM preconditioning stabilises the VE score path’s inputs and denoising residual, while FM changes the target and sampler geometry of the supervised problem. The tuned linear-FM ablation is important here: variance preservation is one helpful conditioning axis, but not the whole story. Linear FM has a less uniform input scale, yet its simple displacement-like velocity target and direct few-step ODE can be easier to exploit once conditional location is residualized. Log- σ sampling controls where tree capacity is spent, the log- σ feature exposes the noise regime to histogram splits, residualization removes predictable conditional location, and Heun ODE sampling is an inference-side efficiency gain. The large-dataset inversion argues against a blanket “SDE is worse” interpretation: per-IQR-bin diagnostics show that stochasticity can improve high-uncertainty tail coverage, while aggregate metrics penalise broader intervals, worse CRPS, and higher latency.

Limitations. All displayed families are tuned per dataset, but this is still finite-budget model selection with unequal search spaces and per-trial costs: the comparison is equal finite-trial budget rather than wall-clock-normalized model selection. We include six probabilistic baselines but not distributional random forests [Cevic et al., 2022]; per-dataset gaps, especially on large datasets, should be read as descriptive rather than high-powered significance estimates. The residualizer is also selected from fixed diagnostics rather than automated jointly with the diffusion objective, which is the most likely reason the published SDE still wins the two largest datasets, where the inner model has enough data to fit the conditional mean directly and the fixed residualizer becomes a bottleneck.

Choosing a variant. The per-dataset pattern supports a simple default rather than *calibration vs. speed* alone. Use headline 5-step VP-FM when sample latency or aggregate normalized CRPS is the binding constraint; it is the fastest displayed headline corner, while the mechanism ablation shows residualized linear FM as the tuned CRPS corner. Prefer score⁺ when interval coverage or PIT uniformity is the reported quantity, since it is the strongest calibrated row at the cost of $\sim 2.4\times$ more sampling time. On the largest datasets, where residualizer capacity bottlenecks, the published SDE remains competitive and full SDE sampling is worth its cost. Stochasticity ($\varepsilon > 0$) is a conditional

211 knob, not a default: select it only when high-uncertainty tail coverage is the target (Appendix K),
212 against conditional-calibration metrics rather than aggregate CRPS.

213 **Related uncertainty methods for trees.** Our baselines cover sample-based and quantile tree
214 learners, but several complementary tree-uncertainty families target different quantities. Epistemic-
215 ensemble boosting such as SGLB [Ustimenko and Prokhorenkova, 2021] uses Langevin dynamics to
216 quantify model uncertainty and is evaluated mainly on NLL/RMSE and out-of-distribution detection
217 rather than the sharpness-and-coverage CRPS regime here. Frequentist inference for boosted trees, e.g.
218 Boulevard [Zhou and Hooker, 2022], yields asymptotic prediction intervals with coverage guarantees
219 but does not provide a full generative predictive law. Conformal wrappers such as conformalized
220 quantile regression [Romano et al., 2019] are orthogonal and could in principle be layered on the
221 deterministic FM or score⁺ outputs to restore finite-sample coverage while preserving the latency
222 advantage; we leave that combination to future work.

223 **Future work.** Useful extensions include wall-clock-normalized tuning, selection criteria for stochas-
224 tic samplers when conditional coverage matters, conformal post-calibration of the deterministic
225 samplers [Romano et al., 2019], multi-output regression, and automatic residualizer selection.

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270 A Theoretical Derivations and Design Consequences

271 **Lemma 1 (Wronskian velocity-to-score identity)** *For a differentiable Gaussian path $y_t =$*
 272 *$\alpha(t)y_0 + \beta(t)z$ with conditional velocity $u_t = \alpha'(t)y_0 + \beta'(t)z$, nonzero Wronskian $W(t) =$*
 273 *$\alpha(t)\beta'(t) - \alpha'(t)\beta(t)$, and $\beta(t) \neq 0$ (away from the data endpoint), the population velocity*
 274 *$v(y_t, x, t) = \mathbb{E}[u_t \mid y_t, x, t]$ implies the conditional score*

$$\nabla \log p_t(y_t \mid x) = \frac{\alpha'(t)y_t - \alpha(t)v(y_t, x, t)}{W(t)\beta(t)}.$$

275 *Proof.* Solving the two linear equations for z gives

$$z = \frac{\alpha(t)u_t - \alpha'(t)y_t}{\alpha(t)\beta'(t) - \alpha'(t)\beta(t)} = \frac{\alpha(t)u_t - \alpha'(t)y_t}{W(t)}.$$

276 Because $p(y_t \mid y_0, x) = \mathcal{N}(\alpha y_0, \beta^2 I)$, the conditional score is $-z/\beta(t)$. Fisher’s identity gives
 277 $\nabla \log p_t(y_t \mid x) = \mathbb{E}[-z/\beta(t) \mid y_t, x, t]$. Substituting the expression for z and using that y_t is fixed
 278 under the conditional expectation yields the claim. Replacing v by v_θ gives Eq. (2). $\square$

279 The identity is not new theory: it specializes the standard velocity/score relations for Gaussian
 280 probability paths and stochastic interpolants [Lipman et al., 2023, Albergo et al., 2023] to the
 281 Wronskian form we use, and we present it only as the bridge that lets a velocity-trained tree model
 282 drive score-based and stochastic-interpolant samplers without retraining.

283 At a fixed t with $\alpha(t) \neq 0$, the same identity inverts to $v(y_t, x, t) = \alpha'(t)y_t/\alpha(t) -$
 284 $W(t)\beta(t)\nabla \log p_t(y_t \mid x)/\alpha(t)$. Thus score and velocity have the same conditional structure as
 285 functions of y_t at that time; the practical differences studied here come from path-dependent scale
 286 variation across t , feature geometry, and sampler dynamics rather than an intrinsic fixed- t smoothness
 287 advantage.

288 **Path-induced feature scale.** If y_0 and z are independent with unit variance after standardization,
 289 then the path feature satisfies

$$\text{Var}(y_t) = \alpha(t)^2 + \beta(t)^2.$$

290 Thus the linear path has $\text{Var}(y_t) = (1-t)^2 + t^2$ and compresses to variance 1/2 at $t = 1/2$, whereas
 291 the trigonometric and VP paths keep $\text{Var}(y_t) = 1$ for all t . This identity is elementary; its role here
 292 is to connect path choice to the empirical distribution from which histogram split candidates are built.

293 **Proposition 1 (Pooled histogram-bin proxy)** *Fix a scalar noised-response feature and a finite par-*
 294 *tition $\mathcal{P} = \{I_1, \dots, I_B\}$. Let $\mathcal{H}(\mathcal{P})$ be the functions that are constant on each I_b , and let*

$$A_t(\mathcal{P}) = \inf_{g \in \mathcal{H}(\mathcal{P})} \mathbb{E}_{p_t}[(f_t(y_t) - g(y_t))^2], \quad f_t(y) = \mathbb{E}[T_t \mid y_t = y, x, t],$$

where T_t is the scalar score, denoising, noise, or velocity target at time t with covariate value x fixed.
Then

$$A_t(\mathcal{P}) \leq \sum_{b=1}^B p_t(I_b) \text{osc}_{I_b}(f_t)^2,$$

where $\text{osc}_{I_b}(f_t) = \sup_{y, y' \in I_b} |f_t(y) - f_t(y')|$. If f_t is L_t -Lipschitz on the bins, then

$$A_t(\mathcal{P}) \leq L_t^2 \sum_{b=1}^B p_t(I_b) |I_b|^2.$$

Moreover, in a compact-support proxy with smooth positive density q and equal- q -mass bins $\mathcal{P}_q$, the high-resolution scaling is

$$A_t(\mathcal{P}_q) \lesssim \frac{L_t^2}{B^2} \int \frac{p_t(y)}{q(y)^2} dy,$$

up to lower-order bin-width terms.

Proof. The best piecewise-constant approximation on a fixed partition cannot be worse than any particular constant choice on each bin. Choosing any value between the infimum and supremum of f_t on I_b gives pointwise squared error at most $\text{osc}_{I_b}(f_t)^2$ on that bin, and integration under p_t gives the first bound. The Lipschitz bound follows from $\text{osc}_{I_b}(f_t) \leq L_t |I_b|$. For the high-resolution proxy, an equal- q -mass bin has $|I_b| \approx 1/(Bq(\xi_b))$ for some $\xi_b \in I_b$, so

$$\sum_b p_t(I_b) |I_b|^2 \approx \sum_b p_t(\xi_b) |I_b|^3 = \frac{1}{B^2} \sum_b \frac{p_t(\xi_b)}{q(\xi_b)^3} \frac{1}{B} \rightarrow \frac{1}{B^2} \int \frac{p_t(y)}{q(y)^2} dy,$$

where the last step is a Riemann sum under the measure with density q . $\square$

Shared-bin mismatch. Proposition 1 is a one-feature proxy for the finite split resolution available to a histogram tree, not a theory of the full boosted LightGBM ensemble. In the benchmark, rows from many noise levels are pooled before tree fitting, so a natural proxy is $q = \bar{p}$ with

$$\bar{p}(y) = \int p(t) p_t(y) dt.$$

The high-resolution term $\int p_t / \bar{p}^2$ is large when $\bar{p}$ allocates little bin density to regions where a particular p_t still places mass. VE paths create exactly this compromise by mixing data-scale low-noise features with expanded high-noise features; a single pooled histogram must spend resolution across both regimes. EDM input preconditioning and VP-FM both reduce this mismatch by keeping the tree feature scale closer to a common range across t . Figure 2(b) illustrates the same effect empirically: the published-style VE target is stable, but the unpreconditioned tree input expands across noise levels, whereas EDM and VP-FM keep the feature scale closer to a shared-bin regime. This is a finite-bin conditioning statement, not a claim that the fixed-time VE regression function is intrinsically rough.

Residualization removes predictable location. Residualization is an exact reparameterization in the infinite-capacity limit: for any fixed $\hat{\mu}(x)$, modelling $r = y - \hat{\mu}(x)$ exactly and returning $y = r + \hat{\mu}(x)$ recovers the same conditional distribution. Its empirical value is finite-model conditioning. In the scalar case, if $\mu(X) = \mathbb{E}[Y | X]$ and $R = Y - \mu(X)$, the law of total variance gives

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mu(X)), \quad \text{Var}(R) = \mathbb{E}[\text{Var}(Y | X)].$$

For vector-valued responses the same statement applies to the trace of the covariance. Thus residualization removes predictable location variation before noised responses are pooled into histogram bins. The large-dataset caveat remains: a fixed residualizer can become a bottleneck once the inner score or velocity model has enough data to represent the conditional mean directly.

Proposition 2 (Endpoint score-error amplification) Let $v_\theta(y_t, x, t) = v(y_t, x, t) + \delta(y_t, x, t)$ be an approximate velocity with pointwise error δ , and let $\hat{s}_\theta, s$ denote the implied and true scores from Lemma 1. Then

$$\hat{s}_\theta(y_t, x, t) - s(y_t, x, t) = -\frac{\alpha(t)}{W(t)\beta(t)} \delta(y_t, x, t),$$

so the pointwise score-error gain is $|\alpha(t)/(W(t)\beta(t))|$. For the linear and trigonometric paths, $\beta(t) \asymp t$ and $W(t)$ is bounded away from zero near $t = 0$, so this gain diverges like $1/t$. For the VP path used here, $\beta(t) \asymp \sqrt{t}$ and $W(t) \asymp 1/\sqrt{t}$, so $W(t)\beta(t)$ is bounded away from zero and the velocity-to-score gain remains bounded.

Proof. Eq. (2) is linear in v , so

$$\hat{s}_\theta - s = \frac{\alpha'(t)y_t - \alpha(t)(v + \delta) - (\alpha'(t)y_t - \alpha(t)v)}{W(t)\beta(t)} = -\frac{\alpha(t)\delta}{W(t)\beta(t)}. \quad \square$$

Endpoint amplification. Proposition 2 therefore predicts that any stochasticity schedule with $\varepsilon(t) \not\rightarrow 0$ as $t \rightarrow 0$ injects recovered-score error into the SDE drift. In Eq. (3), the recovered-score contribution to the drift error is

$$\frac{\varepsilon(t)^2}{2}(\hat{s}_\theta - s) = -\frac{\varepsilon(t)^2\alpha(t)}{2W(t)\beta(t)}\delta.$$

More generally, if $\beta(t) \asymp t^a$, $\varepsilon(t) \asymp t^b$, $\alpha(t) \asymp 1$, and $W(t) \asymp t^{a-1}$ near the data endpoint, this drift-error contribution scales as $t^{2b+1-2a}\delta$. Linear and trigonometric paths have $a = 1$, so constant stochasticity diverges, $\varepsilon(t) \asymp \sqrt{t}$ leaves a non-vanishing endpoint error scale, and $\varepsilon(t) = ct$ makes the recovered-score contribution vanish. The VP path used here has $a = 1/2$, so the velocity-to-score gain is milder, but vanishing stochasticity still suppresses the endpoint contribution and $\varepsilon \equiv 0$ removes it entirely. This is why $\varepsilon(t) = ct$ and the deterministic corner are the two natural schedules in §2.2.

VE population identity. On a VE path $y_t = y_0 + \beta(t)z$ with $\beta'(t) > 0$, normalized velocity, noise prediction, and denoising score recovery imply the same population score:

$$s_t(y_t, x) = -\frac{\mathbb{E}[z \mid y_t, x, t]}{\beta(t)} = \frac{\mathbb{E}[y_0 \mid y_t, x, t] - y_t}{\beta(t)^2}.$$

Indeed, for VE, $\alpha = 1$, $\alpha' = 0$, $u_t = \beta'(t)z$, and $W = \beta'(t)$; Lemma 1 gives $s_t = -\mathbb{E}[z \mid y_t, x, t]/\beta(t)$, and $y_t = y_0 + \beta(t)z$ gives the denoising form. This identity is not a claim that the finite LightGBM problems are identical. EDM fits the preconditioned denoising residual $F = (y_0 - c_{\text{skip}}y_t)/c_{\text{out}}$ and reconstructs $D_\theta = c_{\text{skip}}y_t + c_{\text{out}}F_\theta$, whereas normalized VE-FM would fit z directly. They imply the same score at the optimum but present different targets and feature scalings to LightGBM. This is the useful interpretation of the score⁺-FM tie: both improve the conditioning of the regression problem seen by the tree ensemble, EDM by preconditioning the VE score objective and VP-FM by choosing a better-conditioned probability path.

Table 5: Regression-conditioning view of the main design choices. Here y_0 and z are standardized. “Conditioning issue” refers to the supervised problem fit by LightGBM, not to the population score identity.

Variant	Regressed object	Feature scale	Conditioning issue / fix
VE noise prediction	$-z$	$y_0 + \sigma z$ grows with σ	Unit target, but heteroscedastic input and $1/\sigma$ score amplification.
VE EDM score ⁺	$(y_0 - c_{\text{skip}}y_t)/c_{\text{out}}$	$c_{\text{in}}y_t$ is near unit scale	Input and denoising residual are explicitly preconditioned.
Raw VE-FM	$\beta'(t)z$	$y_0 + \beta z$ grows with β	Velocity target inherits the schedule derivative.
Normalized VE-FM	z	$y_0 + \beta z$ unless also scaled	Population-equivalent to noise prediction after known rescaling.
VP-FM	$\alpha'y_0 + \beta'z$	$\alpha^2 + \beta^2 \approx 1$	Path keeps tree features near the standardized data scale.

B Datasets and Protocol Details

Sources. The ten UCI benchmarks follow the canonical Treeffuser protocol [Beltrán-Velez et al., 2024]: yacht, concrete, energy, wine, kin8nm, naval, power_plant, protein from the UCI repository; california_housing from scikit-learn’s fetch_california_housing;

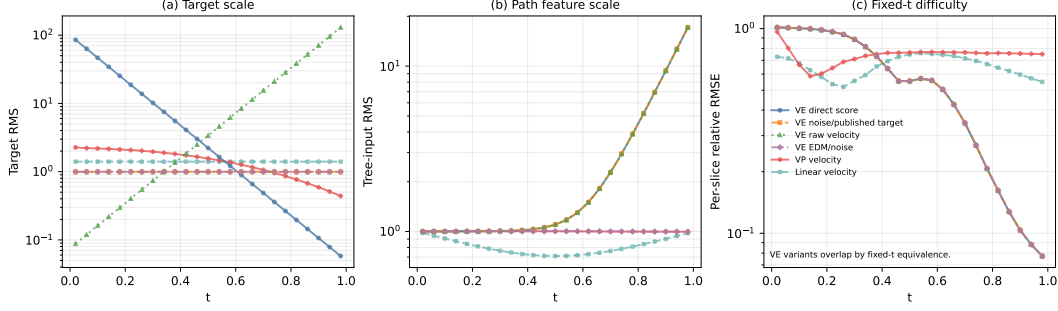


Figure 2: Toy preconditioning diagnostic on a one-dimensional heteroscedastic mixture, generated by `pixi run toy-geometry`. Panel (a) shows that direct VE score prediction and raw VE-FM have target scales that vary by orders of magnitude across t , whereas the published-style VE noise target, EDM/noise prediction, and linear velocity have unit target scale. Panel (b) separates target scaling from input geometry: the published-style VE target is stable but the tree input $y_t = y_0 + \sigma z$ still expands strongly, while EDM rescales the input and VP-FM keeps the path feature near standardized scale. Panel (c) fits a fresh LightGBM at each fixed t ; the VE curves overlap up to numerical noise, confirming that the score/velocity distinction is not intrinsic fixed- t smoothness but cross- t conditioning.

and diabetes from scikit-learn’s `load_diabetes`. The headline results use the six-fold tuning/evaluation protocol from §3; Table 2 reports the post-split training rows for each evaluation fold. Fixed-configuration appendix diagnostics instead use the train/test sizes declared in their YAML configs, sometimes with large datasets subsampled for sweep cost. The only preprocessing is `StandardScaler` standardisation of features and target, fit on the training split and applied to the corresponding test split. `wine` is the union of `wine-quality-red` and `wine-quality-white` with a binary color indicator.

Seed policy. The fold-tuned headline uses the deterministic six-fold manifest with master seed 0. Fixed diagnostic sweeps use the seed lists declared in their configs (usually 0–2 or 0–9), with three independent streams: `data_seed` (offset 0) for the train/test split; `model_seed` (offset 10,000) for the LightGBM regressors; `sampler_seed` (offset 20,000) for stochastic samplers. Variance reduction across seeds therefore reflects only sampling-and-split variability, not boosting initialisation.

Common training settings. In fixed diagnostic sweeps, unless noted, Treeffuser variants use $n_{\text{estimators}} = 3000$, early stopping after 50 rounds, learning rate 0.1, $n_{\text{repeats}} = 30$ (each training point is reused with 30 fresh t samples), and the per-output-dimension LightGBM fitter. In the headline protocol, these and other LightGBM hyperparameters are selected per dataset from the tuning spaces and stored under `benchmarks/results/tuning/best_params/`. Inference draws 200 samples per test point.

Tuning budget. For each dataset/model-family pair, Optuna targets 25 finite trials on the fold-0 train/validation split, with at most twice as many total attempts to allow failed or non-finite trials. The tuning objective is validation CRPS computed from 100 predictive Monte Carlo samples per validation point; this is the number of generated response samples used to estimate CRPS, not a subsample of validation observations. Final evaluation refits the selected configuration on each held-out evaluation fold and draws 200 samples per test point. The three Treeffuser rows share the same six-dimensional LightGBM search surface: `n_estimators`, `learning_rate`, `num_leaves`, `max_depth`, `min_child_samples`, and `subsample`; method-defining choices such as objective, residualization, probability path, and sampler are fixed by row. External baselines are also tuned for 25 finite trials, but their search spaces and per-trial costs differ, so the comparison should be read as equal finite-trial budget rather than wall-clock-normalized model selection. The persisted YAML files record `n_trials_target`, `n_trials_finite`, and `n_trials_failed`. Table 6 summarises the per-family tunable surfaces.

Table 6: Tunable hyperparameter ranges per model family (25 Optuna trials each). Log = log-uniform sampling. Fixed method-defining parameters are not listed.

Family	Tunable parameter	Range	Scale
Treeffuser (all 3 rows)	n_estimators	200–3000	log
	learning_rate	0.001–0.3	log
	num_leaves	15–255	linear
	max_depth	{−1, 4, 6, 8, 10}	categorical
	min_child_samples	5–100	linear
	subsample	0.5–1.0	linear
NGBoost	n_estimators	200–3000	log
	learning_rate	0.001–0.3	log
iBUG	k	10–200	log
	n_estimators	200–2000	log
	learning_rate	0.001–0.3	log
	max_depth	3–10	linear
QReg-LightGBM	quantile_count	24–49	linear
	n_estimators	50–500	log
	learning_rate	0.02–0.2	log
	num_leaves	15–127	linear
CatBoost	iterations	200–3000	log
	learning_rate	0.001–0.3	log
	depth	4–10	linear
Deep ensemble	hidden_size	{64, 128, 256, 512}	categorical
	n_layers	2–5	linear
	learning_rate	0.0001–0.01	log
	max_epochs	{100, 200, 400}	categorical
CARD-style	hidden_size	{64, 128, 256}	categorical
	n_layers	2–5	linear
	learning_rate	0.0001–0.01	log
	max_epochs	{200, 400}	categorical
	diffusion_epochs	{200, 400}	categorical
	n_steps	{50, 100, 200}	categorical

Appendix calibration metrics. Appendix tables use q -MACE for sample-rank calibration: for each held-out response, we compute the fraction of predictive samples above the observed value, sort these ranks, and average their absolute deviation from an evenly spaced uniform grid. Lower q -MACE therefore means the sample predictive distributions have ranks closer to uniform, analogous to the PIT-KS metric in the main table but reported as an error magnitude.

Sampler details. TREEFFUSER-PUBLISHED uses the VESDE reverse-time SDE with the Euler–Maruyama sampler at $n_{\text{steps}} = 50$, matching the published code path. TREEFFUSER-SCORE⁺ uses the Heun probability-flow ODE at $n_{\text{steps}} = 25$ steps under the EDM-style parametrization with $\sigma_{\min} = 0.01$, $\sigma_{\max} = 20$. TREEFFUSER-FM uses the Heun ODE at $n_{\text{steps}} = 5$ on the VP path with $\beta_{\min} = 0.1$, $\beta_{\max} = 20$.

C Claim Scope and Diagnostic Cross-References

Unless otherwise stated, appendix sweeps are fixed-configuration diagnostics used to isolate modelling choices; they are not per-dataset tuned headline comparisons. The experimental pipeline is: fixed diagnostics identify plausible design choices; those choices define the frozen PUBLISHED, SCORE⁺, and FM model families; the frozen families are then tuned per dataset under the fold-0 protocol; and only that tuned protocol supports the headline aggregate performance claims.

The four score-side modifications of §2.1 were validated by smoke and synthetic-core sweeps before being combined in the SCORE⁺ variant. The residualizer-C configuration that the headline uses was selected by the residualizer sweep summarised in Appendix D. Log-sigma time sampling, the log-sigma noise feature, and EDM preconditioning were locked in by the log_sigma_sweep, synthetic_core, and pf_ode_sweep development runs. We treat those runs as model-selection evidence rather than as separately reported tuned comparisons; the quantitative claim in the paper is the combined SCORE⁺ row in Table 1, where all four choices are evaluated together on the full ten-dataset protocol.

Table 7: Scope of the empirical claims. “Tuned” means per-dataset Optuna selection on fold 0 before evaluation on folds 1–5; “fixed” means shared diagnostic settings declared in the corresponding YAML configs.

Claim	Evidence	Tuning status	Role in the paper
SCORE ⁺ and FM improve the Treefuser tradeoff surface over the published row	Table 1 and Table 2	Tuned per dataset	Confirmatory performance comparison among frozen model families
SCORE ⁺ is the most calibrated Treefuser row under this benchmark protocol	Table 1	Tuned per dataset	Headline aggregate calibration claim
VP-FM is the preferred FM path among the tested paths	Table 14	Fixed diagnostic	Mechanism and model-selection evidence; not a claim that VP would win every independently tuned path comparison
Deterministic FM is the recommended aggregate CRPS/latency sampler	Table 13 with Table 18	Fixed diagnostic	Sampler diagnostic; stochasticity remains a conditional-calibration knob
Residualizer-C is a reasonable headline residualizer choice	Tables 8 and 9	Fixed diagnostic	Design choice for the frozen model families; automatic residualizer selection is left as future work
Log- σ sampling, explicit noise-level features, and EDM preconditioning motivate the SCORE ⁺ bundle	Toy diagnostic and development sweeps listed in Table 19	Fixed diagnostic	Mechanism evidence for the bundle; the quantitative performance claim is the tuned combined SCORE ⁺ row

D Residualizer Configuration

D.1 Off vs. Mean Residualization (FM)

We isolate the contribution of mean residualization on the FM side at matched configuration (VP path, uniform t , uniform loss weighting, residualizer-C extra parameters) using a ten-dataset mixed synthetic/real diagnostic suite and ten seeds.

Table 8: FM with mean residualization vs. no residualization (off). Paired differences (mean – off, 100 paired observations); negative is better for all metrics shown.

Sampler	Δ CRPS	Δq -MACE	Δ KS stat	$\Delta cE @90$	paired t on Δ CRPS
ODE @ 5 steps	−0.23	−0.022	−0.034	−0.032	−4.30
SDE @ 25 steps	−0.05	+0.001	+0.003	−0.001	−2.91

The ODE-5 advantage is large and broadly significant across calibration metrics (81/100 paired wins on q -MACE and 81/100 on $|cE|@90$). At SDE-25 the gap on calibration metrics is null (paired $t < 1$); the aggregate CRPS gain is driven by two real-data outliers (yacht +54% relative, concrete +15%) while six of ten datasets actually favour the unresidualized baseline. The mechanism is consistent with the few-step ODE being unable to absorb the conditional mean itself when the velocity net has ≤ 5 Heun updates; with 25 SDE updates the velocity model recovers calibration on its own. Cost: mean residualization adds ≈ 12 s of fit time on these sizes ($\sim 8\times$ vs. no residualization), so the trade is most attractive in low-step regimes.

D.2 Residualizer Variants A–E (FM side)

Five residualizer configurations were swept on a four-dataset, three-seed sub-suite (california_housing, diabetes, energy, protein-5k subsample) under VP-FM-ODE at 5 steps; see Table 9.

Within this fixed diagnostic, configurations C and E sit on the calibration–latency Pareto front. C is the headline choice (best CRPSS, $1.2\times$ faster than E); E edges C on tail coverage ($|cE|@95$ 0.010 vs. 0.011). A separate full-protein diagnostic shows E reclaiming the lead on protein (CRPSS 0.476 vs. C’s 0.449), reinforcing the case that the optimal residualizer is dataset-dependent.

Table 9: FM residualizer diagnostic. “Conf. leaves” is the LightGBM num_leaves; “ $n_{\text{est}}^{(r)}$ ” is the residualizer’s n_estimators; “ES” is early-stopping rounds in the residualizer (off means no ES).

Variant	Description	Leaves	$n_{\text{est}}^{(r)}$	ES	CRPSS $\uparrow$	q -MACE $\downarrow$	cE @95 $\downarrow$	fit s
A	current-baseline	31	100	off	0.521	0.028	0.017	1.82
B	regularised	31	100	30	0.509	0.042	0.018	2.78
C	high capacity	63	300	off	0.523	0.039	0.011	8.03
D	ES moderate	63	500	30	0.516	0.033	0.021	5.91
E	ES + high capacity	63	2000	30	0.522	0.034	0.010	9.81

D.3 Scale-Aware Residualization

The mean_scale residualizer additionally divides residuals by a cross-validated conditional-std estimate $\hat{\sigma}(x)$. On the ten-dataset mixed synthetic/real diagnostic suite with ten seeds:

Table 10: Mean vs. mean-scale residualization on FM at matched configuration. Higher CRPSS is better; lower CRPS, DSS, q -MACE are better.

Variant	CRPSS $\uparrow$	CRPS $\downarrow$	DSS $\downarrow$	q -MACE $\downarrow$
VP-FM-ODE, mean	0.519	0.652	+0.314	0.031
VP-FM-ODE, mean-scale	0.489	0.724	+0.665	0.038
VP-FM-SDE, mean	0.517	0.657	+0.380	0.040
VP-FM-SDE, mean-scale	0.475	0.750	+0.676	0.049

Scale-aware residualization is uniformly worse: +11% CRPS on the ODE path, +14% on the SDE path, with DSS more than doubling. The per- x scale estimator is itself noisy in these finite-sample regimes; dividing by it injects more variance into the velocity target than it removes. mean is therefore the only residualizer carried into the headline.

E Time Sampling and Loss Weighting

E.1 Time Sampling

Table 11: FM time-sampling sweep on fixed UCI train/test splits, 10 datasets $\times$ 3 seeds, matched residualizer-C, VP path. “uniform” is the headline choice with an endpoint anchor at $t = 1$ at probability 0.05.

Variant	CRPSS $\uparrow$	DSS $\downarrow$	q -MACE $\downarrow$	KS $p > .05 \uparrow$	samp s
VP-FM-ODE uniform	0.6564	−0.154	0.040	0.47	3.63
VP-FM-ODE uniform anchor16	0.6560	−0.171	0.040	0.53	3.84
VP-FM-ODE logbeta	0.6556	−0.175	0.035	0.77	3.74
VP-FM-ODE logSNR	0.6538	−0.135	0.042	0.40	3.71
score ⁺ uniform- t	0.6556	−0.210	0.038	0.50	6.95
score ⁺ log- σt	0.6565	−0.169	0.038	0.70	6.58

For FM, log-beta and log-SNR sampling do not move CRPS by more than 0.4% relative to uniform with endpoint anchor; the smoother density shape gains +5% on KS pass rate and −13% on q -MACE but the effect is within seed variance for CRPSS. The headline keeps uniform- t for parsimony; log-beta is a viable substitute when KS uniformity is the priority. For the score model, log-sigma sampling clearly helps PIT uniformity (KS pass 0.50 $\rightarrow$ 0.70) at the cost of slightly worse DSS; we keep log-sigma in the SCORE⁺ headline because the KS gain is the more interpretable calibration win.

E.2 Loss Weighting (min-SNR- γ)

Min-SNR- γ is a sharpness-versus-tail trade-off. $\gamma = 1$ hurts CRPS by 0.8–1.3% across both samplers (Wilcoxon $p \leq 0.003$ for both); $\gamma = 5$ is statistically indistinguishable from uniform weighting on

Table 12: Min-SNR- γ loss weighting on FM, fixed UCI train/test splits, 10 datasets $\times$ 3 seeds. $\gamma = 5$ collapses to uniform on most of the support; $\gamma = 1$ amplifies the late-time loss most aggressively.

Variant	CRPSS $\uparrow$	CRPS $\downarrow$	DSS $\downarrow$	q -MACE $\downarrow$	KS $p > .05 \uparrow$
VP-FM-ODE uniform- w	0.6564	4.097	-0.159	0.040	0.47
VP-FM-ODE min-SNR-5	0.6564	4.119	-0.167	0.037	0.60
VP-FM-ODE min-SNR-1	0.6516	4.223	-0.104	0.044	0.37
VP-FM-SDE uniform- w	0.6540	4.114	-0.080	0.051	0.30
VP-FM-SDE min-SNR-5	0.6538	4.138	-0.085	0.050	0.30
VP-FM-SDE min-SNR-1	0.6485	4.239	-0.046	0.057	0.23

CRPS, but improves PIT KS pass rate on the ODE path. We keep uniform weighting in the FM headline.

F Stochasticity (Velocity-Noise) Sweeps

F.1 Stochasticity Strength

We sweep velocity stochasticity on the VP path under linear and $\sqrt{t}$ schedules with residualizer-C, plus one residualizer-E check; results in Table 13 are averages over fixed UCI train/test splits, 10 datasets $\times$ 3 seeds at 25 SDE steps.

Table 13: Stochasticity sweep. The $\varepsilon = 0$ corner is the deterministic FM recipe (Heun ODE @ 5 steps); SDE rows use 25 steps. Sample time is omitted because the ODE and SDE rows use different benchmark protocols.

Variant	CRPSS $\uparrow$	CRPS $\downarrow$	cE @90 $\downarrow$	cE @95 $\downarrow$
$\varepsilon = 0$ headline (ODE)	0.669	—	0.024	0.017
$\varepsilon = 0.25$ linear, res-C	0.656	4.099	0.028	0.015
$\varepsilon = 0.50$ linear, res-C	0.655	4.104	0.036	0.017
$\varepsilon = 1.00$ linear, res-C	0.654	4.114	0.047	0.022
$\varepsilon = 0.50$ sqrt, res-C	0.655	4.101	0.035	0.017
$\varepsilon = 1.00$ sqrt, res-C	0.655	4.107	0.043	0.020
$\varepsilon = 1.00$ linear, res-E	0.651	4.128	0.048	0.024

In this fixed diagnostic sweep, no tested SDE setting Pareto-dominates $\varepsilon = 0$. The closest contender ($\varepsilon = 0.25$ linear) matches |cE|@95 (0.015 vs. 0.017) but is strictly worse on CRPSS and |cE|@90, and pays 25 SDE steps vs. 5 ODE steps.

F.2 Schedule Shape

The schedule $\varepsilon(t) = ct$ (linear) is the headline form. The $\sqrt{t}$ alternative was tested for parity at $\varepsilon \in \{0.5, 1.0\}$ and produces statistically indistinguishable CRPSS (Table 13). Both schedules vanish at the data endpoint, which suppresses recovered-score error in the stochastic drift (Eq. (2) and Proposition 2); neither shape helps once ε is small.

G Path Choice

Table 14: Flow-path diagnostic under the 5-step deterministic ODE sampler: linear vs. trigonometric vs. variance-preserving, 8 datasets $\times$ 3 seeds, matched residualizer-C. Lower is better for all columns; bold is best per column.

Path	CRPS $\downarrow$	q -MACE $\downarrow$	cE @95 $\downarrow$	samp s
Linear	4.324	0.026	0.032	0.49
Trigonometric	4.306	0.028	0.028	0.43
Variance-preserving (VP)	4.294	0.028	0.028	0.46

VP is best on CRPS in this fixed few-step ODE diagnostic, while trig and VP both reduce 95% coverage error relative to the linear path. The advantage tracks variance preservation: VP and trig satisfy $\alpha^2 + \beta^2 \approx 1$, keeping y_t at roughly unit scale across t and stabilising the tree regressor’s feature space. The tuned mechanism ablation in Table 4, however, shows that residualized linear FM can beat residualized VP-FM on aggregate CRPSS/rel-CRPS once the LightGBM hyperparameters are selected per dataset. We therefore read path choice as a finite-model tradeoff: VP gives stable feature scale and the fastest displayed FM sampler, while linear FM can have a simpler displacement-like velocity target that wins CRPS after residualization and tuning.

H Probabilistic-Family Baselines

We compare the headline Treeffuser variants against six external probabilistic regressors (Table 15). Hyperparameters for every displayed family are selected per dataset on the same fold-0 tuning split, then evaluated on folds 1–5. The CARD row uses a compact benchmark adapter with the same two-stage conditional-mean plus conditional-diffusion structure as CARD [Han et al., 2022], so conclusions involving CARD should be read as comparisons to this adapter rather than to a fully optimized CARD pipeline.

Table 15: Probabilistic baselines under the tuning/evaluation protocol. rel-CRPS is normalized by the best displayed variant on each dataset. Sample-time excludes fitting. The iBUG DSS is dominated by a single wine fold where the model assigns near-zero predictive variance to a mispredicted test point.

Model	CRPSS $\uparrow$	rel-CRPS $\downarrow$	CRPS $\downarrow$	DSS $\downarrow$	q -MACE $\downarrow$	cE @95 $\downarrow$	samp s
Treeffuser-score ⁺ (ours)	0.681	1.131	4.057	−0.214	0.028	0.015	34.12
Treeffuser-FM-fast (ours)	0.680	1.117	4.121	− 0.218	0.034	0.024	14.41
Treeffuser-published	0.670	1.242	4.279	1.358	0.044	0.023	221.35
Quantile-regression LightGBM	0.668	1.459	4.026	0.016	0.033	0.044	0.30
Deep ensemble	0.654	1.519	4.008	0.090	0.050	0.018	0.03
iBUG (XGBoost)	0.649	1.471	4.193	1.85×10^7	0.069	0.065	2.89
CatBoost (uncertainty)	0.638	1.409	4.045	0.272	0.045	0.066	0.01
CARD-style diffusion	0.631	1.632	4.838	0.436	0.051	0.043	42.84
NGBoost (Gaussian)	0.601	1.550	4.299	1.957	0.052	0.101	0.04

Under this per-dataset tuning protocol, the Treeffuser variants lead on aggregate CRPSS, rel-CRPS, and coverage metrics. The external baselines are still important: deep ensembles have the best mean raw CRPS, though this scale-sensitive average is less informative about cross-dataset dominance than CRPSS or rel-CRPS, and the per-dataset winners include deep ensembles on diabetes and kin8nm, CARD on naval, and quantile-regression LightGBM on protein. Their aggregate rel-CRPS remains worse, indicating that no single tuned external family transfers as evenly across the benchmark suite as the Treeffuser variants.

I Extended Per-Dataset Results

Table 16 reports CRPS, |cE|@90, and sample-generation time on each of the ten benchmarks for the three headline Treeffuser variants, averaged over the five evaluation folds. The “best margin” column gives the gap between the winning variant and the second-best Treeffuser variant in raw CRPS. After per-dataset tuning, the margins are tighter: the main result is no longer a few dramatic wins, but a more balanced Pareto surface between published SDE sampling, score-side conditioning, and FM speed.

FM is the fastest Treeffuser sampler on 9/10 datasets and is especially valuable at scale: on protein the speedup is $19.3\times$ over PUBLISHED and $2.0\times$ over SCORE⁺; on california_housing the corresponding speedups are $31.6\times$ and $2.7\times$. On |cE|@90, however, SCORE⁺ leads on 6/10 datasets and FM on 3/10, which is why Table 1 separates normalized-CRPS and calibration conclusions. Under this evaluation protocol, the per-dataset CRPS ranking reproduces the headline pattern: FM has the best average normalized CRPS, score⁺ is the more reliable calibration row, and the published SDE remains best on wine, california housing, and protein.

Table 16: Extended per-dataset Treeffuser results, ordered roughly by evaluation-fold training size. “best margin” is winner-over-second-best in raw CRPS; the bold cell in each block is the per-metric winner on that row. Numbers come from `benchmarks/results/tuning/eval/*.jsonl`.

Dataset	best margin	CRPS ↓			cE @90 ↓			sample s		
		Pub.	Score ⁺	FM	Pub.	Score ⁺	FM	Pub.	Score ⁺	FM
yacht	FM +0.4%	0.438	0.284	0.283	0.092	0.025	0.081	0.3	0.3	0.1
diabetes	Sco ⁺ +1.9%	35.682	34.025	34.660	0.079	0.029	0.038	2.4	0.5	0.2
energy	FM +5.2%	0.252	0.214	0.204	0.081	0.034	0.022	2.9	0.8	0.6
concrete	Sco ⁺ +1.5%	2.457	2.126	2.158	0.034	0.025	0.023	3.0	4.8	0.8
wine	Pub +5.9%	0.300	0.319	0.318	0.037	0.006	0.016	100.4	37.0	11.9
kin8nm	FM +4.0%	0.067	0.058	0.055	0.012	0.004	0.023	22.0	20.2	9.2
power	FM +2.4%	1.671	1.572	1.535	0.013	0.004	0.006	14.6	6.7	7.6
naval	FM +3.2%	0.000218	0.000220	0.000211	0.078	0.065	0.061	207.4	91.2	29.5
cali. housing	Pub +3.6%	0.197	0.204	0.206	0.010	0.005	0.007	605.9	52.6	19.2
protein	Pub +2.7%	1.722	1.769	1.796	0.005	0.005	0.007	1254.7	127.0	64.9

J Robustness by Dataset Size

Table 17 groups the tuned evaluation artifacts by post-split training rows: small datasets have at most 1,000 rows, medium datasets have 1,001–10,000 rows, and large datasets have more than 10,000 rows. The grouping sharpens the mixed-winner story. Score⁺ and FM lead the small and medium groups on CRPSS, while the two largest datasets favor the published SDE and quantile-regression LightGBM. Thus the aggregate Treeffuser result is not driven by a single data-scale regime, but neither does it erase the large-dataset inversion discussed in the limitations.

Table 17: Dataset-size robustness summary. Entries are mean CRPSS within each size group after first averaging the five evaluation folds per dataset.

Model	Small (4)	Medium (4)	Large (2)
Treeffuser-published	0.709	0.670	0.591
Treeffuser-score ⁺	0.735	0.678	0.578
Treeffuser-FM	0.731	0.683	0.573
QReg-LightGBM	0.711	0.666	0.587
CatBoost-unc.	0.724	0.612	0.517
NGBoost	0.718	0.550	0.468
Deep ensemble	0.699	0.672	0.529
iBUG	0.700	0.637	0.571
CARD-style diffusion	0.632	0.667	0.555

The rank diagnostics in Table 3 give the same guarded conclusion. The Friedman test detects differences among the nine families, but Nemenyi post-hoc comparisons at $\alpha = 0.05$ only separate score⁺ from NGBoost and FM from NGBoost/CARD-style diffusion. They do not separate FM, score⁺, the published SDE, QReg-LightGBM, or deep ensembles from one another, so we use these ranks as robustness context rather than as a stronger significance claim.

K Conditional Tail Calibration Diagnostic

The large-dataset inversion in Table 2 admits two plausible mechanisms: the fixed residualizer may be suboptimal at high n , and stochastic samplers may improve conditional tail calibration in high-uncertainty regions. Table 18 reports direct 95% coverage on `california_housing` and `protein`, including the highest predicted-IQR bin.

The diagnostic supports a tradeoff rather than a one-sided conclusion. On `california_housing`, the deterministic FM rows under-cover the high-IQR bin (0.900), while adding FM stochasticity lifts this coverage to 0.938–0.944; the price is marginal over-coverage (0.967–0.971) and no gain over the published SDE on CRPS or IQR-MACE. On `protein`, residualizer-E with the deterministic ODE is already close to the nominal target and remains best on CRPS and IQR-MACE, whereas the residualizer-E SDE rows over-cover both marginally and in the high-IQR bin. Residualizer-C stochasticity is directionally useful for coverage, especially with the square-root schedule, but gives up CRPS. Thus the safest interpretation is not that SDE samplers are dominated, but that their benefit

Table 18: High-uncertainty tail diagnostic on the two largest datasets. Cov@95 is marginal empirical coverage; Top-IQR Cov@95 is empirical coverage in the highest predicted-IQR quintile; both target 0.95. IQR-MACE is mean $|cE|$ @95 across predicted-IQR quintiles, so lower is better. SDE-linear uses a linear stochasticity schedule; SDE- $\sqrt{t}$ uses the square-root schedule. All rows use the full training and test splits in a separate three-seed diagnostic.

Dataset	Variant	CRPS	Cov@95	IQR-MACE	Top-IQR Cov@95
cali. housing	Published SDE	0.207	0.958	0.015	0.931
	FM ODE-C	0.214	0.947	0.021	0.900
	FM ODE-E	0.213	0.947	0.021	0.900
	FM SDE-C linear	0.215	0.969	0.023	0.939
	FM SDE-C $\sqrt{t}$	0.215	0.967	0.022	0.938
	FM SDE-E linear	0.213	0.971	0.023	0.944
protein	FM SDE-E $\sqrt{t}$	0.213	0.969	0.023	0.944
	Published SDE	1.787	0.944	0.018	0.925
	FM ODE-C	1.860	0.933	0.025	0.923
	FM ODE-E	1.771	0.949	0.008	0.943
	FM SDE-C linear	1.859	0.959	0.011	0.960
	FM SDE-C $\sqrt{t}$	1.856	0.956	0.012	0.954
	FM SDE-E linear	1.779	0.974	0.024	0.973
	FM SDE-E $\sqrt{t}$	1.776	0.971	0.021	0.971

531 is conditional and must be selected against conditional-calibration metrics rather than headline CRPS
532 alone.

533 L Reproducibility

534 The submission includes an anonymized frozen repository via anonymous.4open.science. The
535 repository contains the implementation in src/, the benchmark harness in benchmarks/, the pa-
536 per sources in paper/, and the pixi.toml/pixi.lock environment used for the reported runs.
537 All sweeps in this paper are driven by YAML configs under benchmarks/configs/. Each re-
538 sult file is a JSON Lines document recording the full hyperparameter set, random seeds, the git
539 commit hash of the model code, a hash of the Treeffuser source tree, and every reported met-
540 ric on a per-dataset/per-fold row. The headline tuning runs persist their selected configurations
541 under benchmarks/results/tuning/best_params/ and their five-fold evaluation rows under
542 benchmarks/results/tuning/eval/. The configs and result artifacts that produced the tables
543 are listed alongside in Table 19; the paper PDF is rebuilt with the paper pixi task.

Table 19: Provenance: appendix tables and the config / result file each was produced from.

Table	Config	Result file
8	fm_off_vs_mean_sweep.yaml	fm_off_vs_mean_sweep_*.jsonl
5	analytic summary	-
9	residualizer_fm_sweep.yaml	residualizer_fm_sweep_*.jsonl
10	fm_mean_scale_sweep.yaml	fm_mean_scale_sweep_*.jsonl
11	fm_t_sampling_sweep.yaml	fm_t_sampling_sweep_*.jsonl
12	fm_loss_weighting_sweep.yaml	fm_loss_weighting_sweep_*.jsonl
13	sde_lower_stoch_sweep.yaml, stochasticity_schedule_sweep.yaml	sde_lower_stoch_sweep_*.jsonl, stochasticity_schedule_sweep_*.jsonl
14	ode_vs_sde_path_ablation.yaml	ode_vs_sde_path_ablation_*.jsonl
15, 2, 16, 1	tuning_manifest.yaml	benchmarks/results/tuning/eval/*.jsonl
3, 17	benchmarks/scripts/summarize_robustness.py	benchmarks/results/selected/robustness_summary.md
4	mechanism_ablation_manifest.yaml	benchmarks/results/mechanism_ablation/eval/*.jsonl
1	benchmarks/scripts/run_sampler_cost_sweep.py, benchmarks/scripts/summarize_sampler_cost_sweep.py	benchmarks/results/sampler_cost_sweep/eval/sampler_cost_sweep.jsonl
18	paper_real_data_v2.yaml, tail_calibration_full.yaml	paper_real_data_v2_full.jsonl, tail_calibration_full.jsonl
6	benchmarks/tuning/search_spaces.py	-

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